



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Pathology of Thrombosis

Principles and Practices of Osteopathic Medicine I

Maria M. Plummer, M.D.

Associate Professor

Department of Clinical Specialties

mplummer@nyit.edu



Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Learning objectives

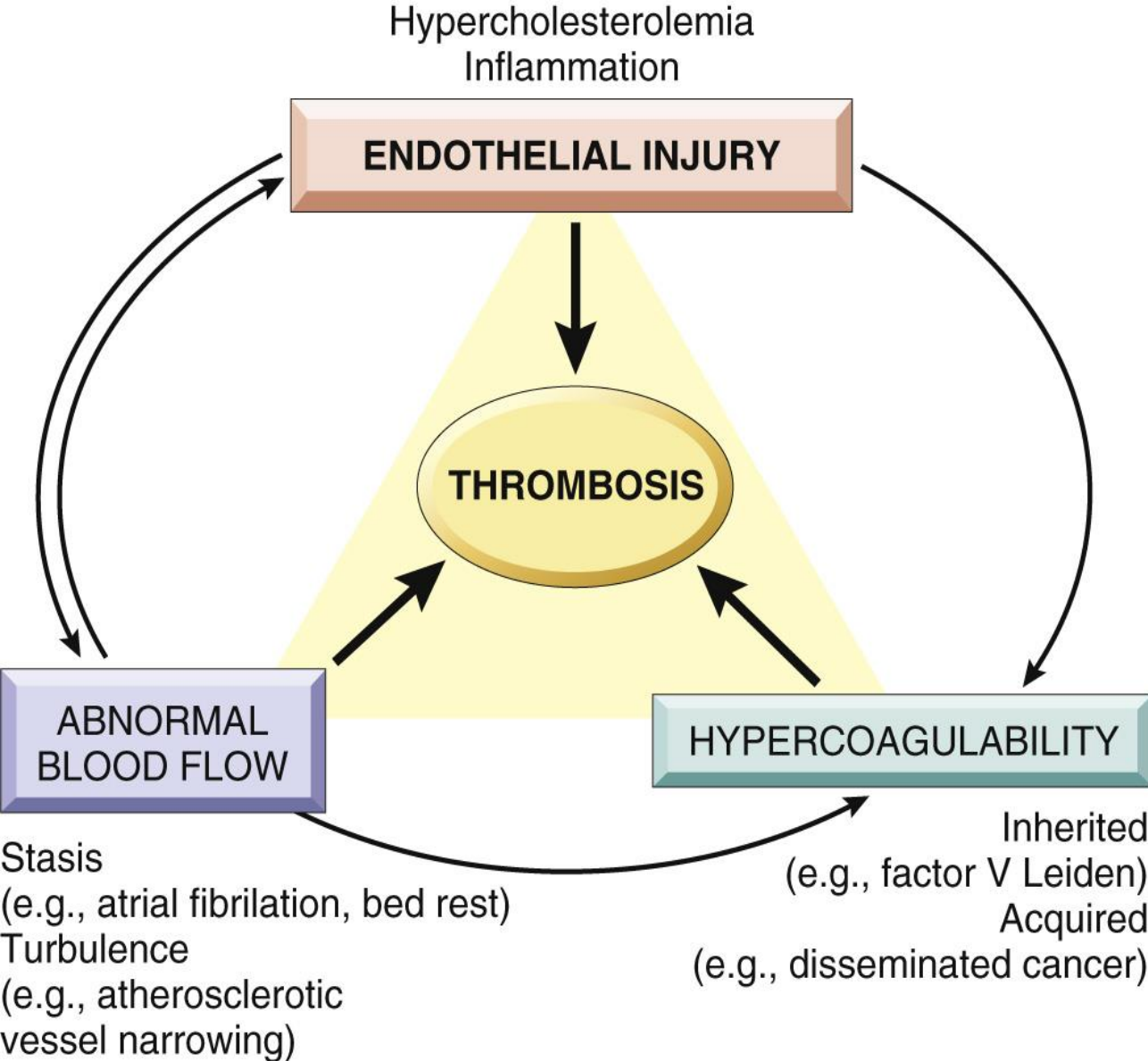
1. After describing the **morphology of thrombi (including postmortem clots)**, discuss and compare the primary thrombotic diseases which include **Factor V Leiden, Prothrombotic Gene mutation, Homocysteinemia, Anti-thrombin III deficiency, and Protein S and Protein C deficiency**
2. Describe and discuss secondary thrombotic disorders which include **malignancies, etc.** and heparin induced thrombocytopenia, **and anti-phospholipid syndrome**
3. Describe, discuss, compare the **fate of thrombi, venous v. arterial and cardiac thrombi, emboli including pulmonary emboli, systemic emboli, and paradoxical emboli**
4. Describe, discuss and compare other various emboli including **fat emboli, air emboli, and amniotic fluid emboli**

OVERVIEW

- Virchow's triad (endothelial injury, stasis, hypercoagulability)
- Characteristics of thrombi (venous, arterial, mural, postmortem)
- Primary (hereditary) thrombophilias
- Secondary (acquired) thrombophilias
- Fate of thrombus
- Embolism
 - Thromboembolism (pulmonary and systemic),
 - Atheroembolism
 - Fat/marrow embolism
 - Gas embolism
 - Amniotic fluid embolism

Thrombosis Definition and Epidemiology

- Development of blood clot within vessel: obstructs natural blood flow, leads to clinical sequelae
- Mortality rate depends on location and acuity of thrombosis
- MI and CVAs account for highest proportion of thrombosis associated deaths in the US



The Virchow triad in thrombosis: Endothelial integrity most important factor. **Injury to endothelial cells** changes local blood flow and affects coagulability

Abnormal blood flow (stasis or turbulence), can cause endothelial injury.

3 Major risk factors for Thrombosis

Virchow's Triad

1. Endothelial Injury

- Atherosclerosis / Hypercholesterolemia / Inflammation
- HTN / Vasculitis / Diabetes
- Toxins (Cigarette smoke) / Elevated Homocysteine

2. Abnormal Blood Flow

- Stasis / Turbulence

3. Hypercoagulable State

- Primary (genetic) or Secondary (acquired) disorders

Endothelial Injury

- **Severe endothelial injury exposes vWF and tissue factor: may trigger thrombosis**
- **Inflammation** promotes thrombosis: shifts gene expression in endothelium to “prothrombotic.” (i.e., **endothelial activation or dysfunction**) – causes include:
 - physical injury
 - infectious agents
 - abnormal blood flow
 - inflammatory mediators
 - metabolic abnormalities (e.g., hypercholesterolemia or homocystinemia)
 - toxins absorbed from cigarette smoke
- **Procoagulant changes:** e.g.,: endothelial cells activated by inflammatory cytokines, downregulate expression of thrombomodulin **procoagulant and pro-inflammatory actions of thrombin enhanced**
- **Antifibrinolytic effects:** activated endothelial cells secrete plasminogen activator inhibitors (PAIs) (limit fibrinolysis and downregulate expression of t-PA) – favors development of thrombi

Turbulence (Abnormal blood flow)

- **Turbulence contributes to arterial and cardiac thrombosis by causing endothelial injury/ dysfunction and forming countercurrents (contribute to stasis)**
- **Turbulence and stasis:**
 - Promote endothelial activation, enhancing procoagulant activity
 - Disrupt laminar flow- bring platelets into contact with endothelium
 - Prevent washout and dilution of activated clotting factors by fresh flowing blood and inflow of clotting factor inhibitors
- **Examples:**
 - **Ulcerated atherosclerotic plaques** expose subendothelial vWF and tissue factor, (also cause turbulence)
 - **Aneurysms** result in local stasis
 - **Acute myocardial infarction** - areas of noncontractile myocardium associated with stasis and flow abnormalities (promote formation of cardiac mural thrombi)

Hypercoagulability

- Abnormal tendency of blood to clot, usually caused by changes in coagulation factors
- Important in venous thrombosis
- Divided into primary (genetic) and secondary (acquired) disorders
- Primary include:
 - **Factor V Leiden**
 - **Prothrombin gene mutation**
 - **Elevated levels of homocysteine**
 - Rare inherited causes of primary hypercoagulability: **deficiencies of anticoagulants (e.g., antithrombin III, protein C, or protein S)**; present with venous thrombosis and recurrent thromboembolism beginning in adolescence or early adulthood

Pathology of Thrombosis

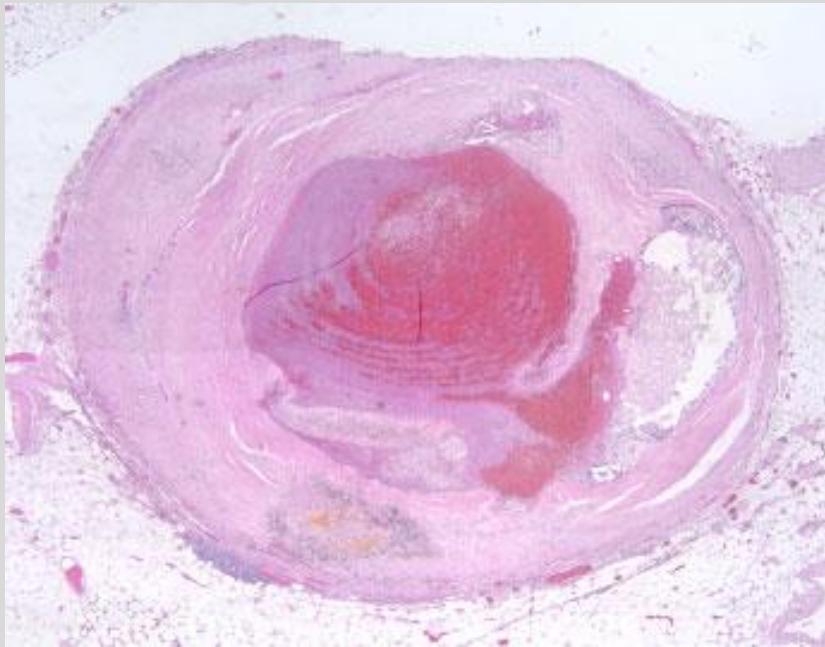
Thrombi

- Size and shape vary, depend on site and underlying cause; obstruct arteries or veins, or embolize
- Gross and microscopic finding: **lines of Zahn**
 - pale platelet and fibrin deposits alternating with darker red cell-rich layers
 - **indicate thrombus formed in flowing blood (antemortem)**; versus bland, non-laminated, postmortem clots
- Focally attached to underlying vascular surface
- Arterial thrombi (atherosclerotic plaques, MI) tend to grow retrograde, and venous thrombi (DVTs) extend in direction of blood flow (**both propagate toward heart**)

Pathology of Thrombosis

Lines of Zahn

- Pale - Platelet and Fibrin
- Dark - RBC's and Fibrin

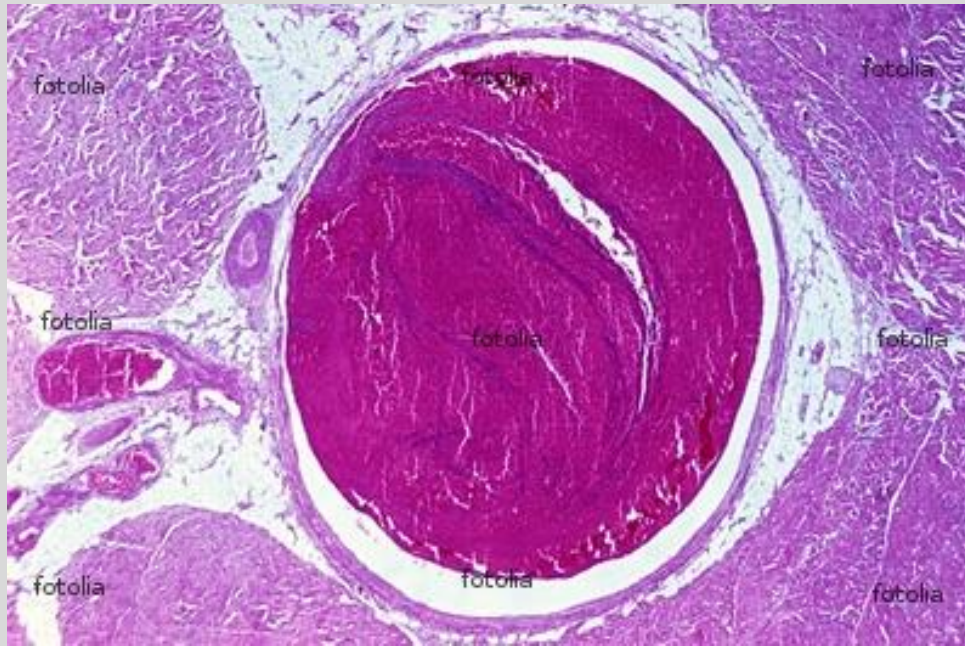


Lines of zahn



Lines of Zahn – Venous Thrombus (usually smaller veins, less frequent) –

- Stasis leads to “Red Thrombi” – More RBCs and less platelets



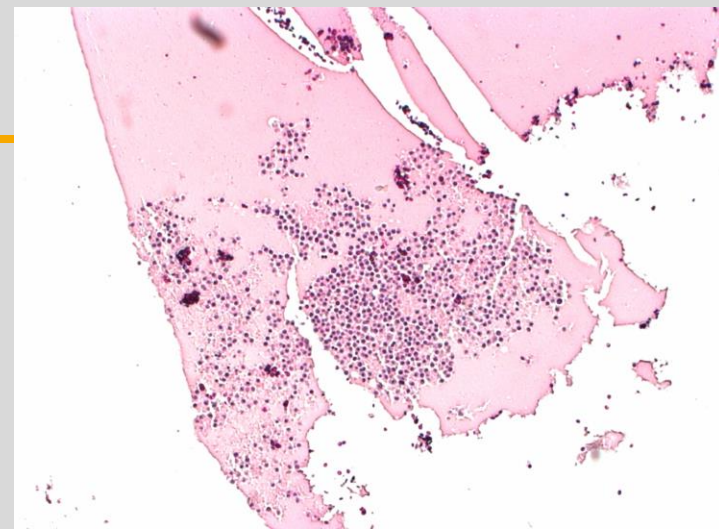
pathology.washington.edu

<https://www.studyblue.com/notes/note/n/pathlab3-hemodynamic/deck/11811180>

Pathology of Thrombosis

Post Mortem Clot: non-laminated (no lines of Zahn)

- Gelatinous with dark-red portion (where RBCs settle due to gravity) and yellow portion
- Distinguished from thrombi in living individuals due to ABSENCE OF LINES OF ZAHN
 - Flowing blood is a requirement for these lines



Gelatinous with dark red dependent portion and yellow "chicken fat" upper portion

Pathology of Thrombosis

- **Venous thrombi**
 - can cause painful congestion and edema distal to obstruction
 - **may embolize to lungs; almost always occlusive**
 - Form long luminal cast – form in sluggish venous circulation - **usually occur at sites of stasis**
 - aka **red thrombi**; firm, focally attached to vessel wall, contain lines of Zahn
 - **veins of lower extremities most commonly involved (90% of cases)**
- **Arterial thrombi**
 - can also embolize; cause downstream infarctions; often occlusive
 - most common sites: **coronary, cerebral**, and femoral arteries (decreasing order of frequency)
 - **Arterial or cardiac thrombi usually begin at sites of turbulence or endothelial injury**
 - Thrombi in heart chambers **aka mural thrombi, esp. seen in:**
 - abnormal myocardial contraction (e.g., arrhythmias, dilated cardiomyopathy, myocardial infarction)
 - endomyocardial injury (myocarditis or catheter trauma)
- **Postmortem clots** - gelatinous with dark-red dependent portion (red cells settled by gravity) and yellow “chicken fat” upper

Venous Thrombosis (Phlebothrombosis)

- **Most occur in superficial or deep veins of leg**
- **Superficial venous thrombi usually occur in saphenous veins in setting of varicosities; can cause local congestion, swelling, pain, and tenderness, rarely embolize**
- **Deep venous thrombus involving large leg veins (e.g., popliteal, femoral, and iliac)— more often embolize to lungs; may cause pulmonary infarction**
- DVTs may cause local pain and edema due to venous obstruction, but because of venous collateral channels, **about 50% of DVTs asymptomatic; recognized only after embolization**
- Lower extremity DVTs often associated with hypercoagulable states; common predisposing factors - **bed rest and immobilization, congestive heart failure; trauma, surgery, and burns** cause immobilization and also vascular insults

Clinical Features



Venous Thrombosis (Stasis)

- Immobilization (Air Travel)
- Bed rest
- Congestive heart failure
- Trauma / Surgery
- Obesity
- Pregnancy: **Thrombotic diathesis of pregnancy**: decreased venous return from leg veins and systemic hypercoagulability secondary to hormonal changes
- Cancer **Tumor-associated inflammation** and coagulation, as well as **procoagulants** (e.g., mucin) **released from tumor cells**: contribute to increased risk of thromboembolism in disseminated cancers (e.g., **migratory thrombophlebitis or Trousseau syndrome**)

Arterial and Cardiac Thrombosis

- **Atherosclerosis**: major cause of arterial thromboses due to loss of endothelial integrity and abnormal blood flow
- **Myocardial infarction** can predispose to cardiac mural thrombi by causing dyskinetic myocardial contraction and endocardial injury
- **Rheumatic heart disease** may cause atrial mural thrombi by causing atrial dilation and fibrillation
- Both cardiac and aortic mural thrombi prone to embolization to any tissue but **especially the brain, kidneys, and spleen**

Primary (Genetic)

Robbins and Cotran, Pathological Basis of Disease, 10th ed., 2020, Ch.4, Table 4.2 Hypercoaguable states

Common

- **Factor V mutation: factor V Leiden** (Arg to Gln substitution in amino acid residue 506 leading to resistance to activated protein C)
- **Prothrombin mutation** (G20210A noncoding sequence variant leading to increased prothrombin levels)
- Increased levels of factors VIII, IX, XI, or fibrinogen (genetics unknown)

Rare

- **Antithrombin III deficiency**
- **Protein C deficiency**
- **Protein S deficiency**

Very Rare

- Fibrinolysis defects
- Homozygous homocystinuria (deficiency of cystathione β -synthetase)

Secondary (Acquired)

Strong Risk Factors for Thrombosis

- **Prolonged bed rest or immobilization**
- Myocardial infarction
- Atrial fibrillation
- Tissue injury (surgery, fracture, burn)
- **Cancer**
- Prosthetic cardiac valves
- Disseminated intravascular coagulation
- **Heparin-induced thrombocytopenia**
- **Antiphospholipid antibody syndrome**

Other Risk Factors for Thrombosis

- Cardiomyopathy
- Nephrotic syndrome
- **Hyperestrogenic states (pregnancy and postpartum)**
- **Oral contraceptive use**
- Sickle cell anemia
- **Smoking**

Pathology of Thrombosis

Classic presentation – recurrent DVTs or DVT at young age

Primary (Genetic)

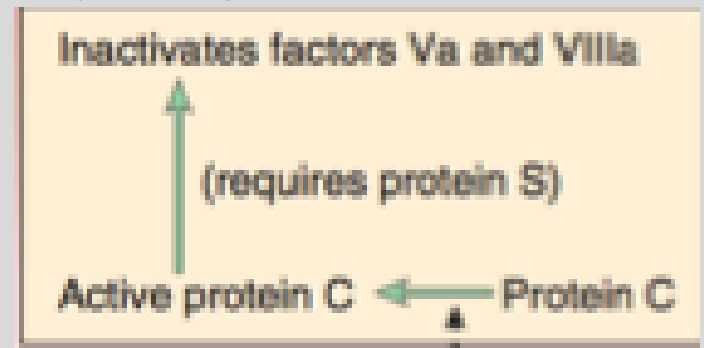
- Factor V Leiden
- Prothrombin Gene Mutation
- Homocysteinemia
- Anti-Thrombin III Deficiency
- Protein C Deficiency
- Protein S Deficiency

Secondary (Acquired)

- Heparin Induced Thrombocytopenia (HIT)
- Anti-Phospholipid Syndrome (APLS)
- Others: Prolonged bed rest, MI, Afib, Cancer, DIC, Post-Op, Prosthetic cardiac valves

FACTOR V LEIDEN

- **Most common hereditary thrombosis syndrome**, 2% to 15% of Caucasians are carriers
- **Autosomal Dominant** with incomplete penetrance
- Single nucleotide mutation makes **Factor V resistant to cleavage** and inactivation **by protein C** (loss of antithrombotic regulator)
- Heterozygotes: 5X increased risk venous thrombosis, homozygotes 50X increase
- Factor V Leiden heterozygote (usually asymptomatic) may develop DVTs when combined with inactivity (e.g., prolonged airplane travel)
- Mutation frequency near 60% in patients with recurrent DVTs
- Patients < 50 years old with thrombotic events should be considered for any clotting disorder, especially Factor V Leiden — even when acquired risk factors are present
- Diagnosis: Activated Protein C (APC) Assay (failure to prolong PTT with addition of purified protein C) or **genetic testing**



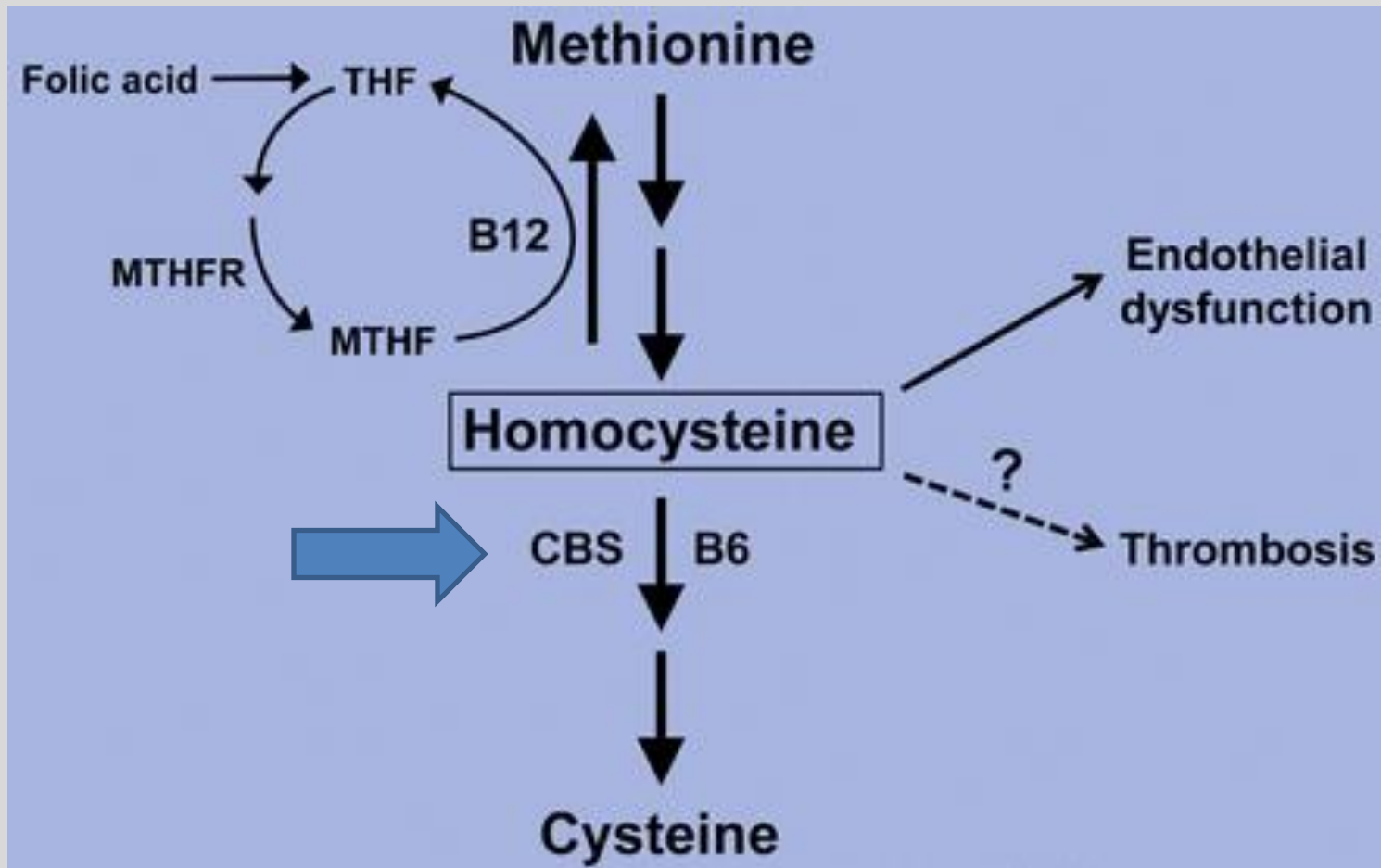
PROTHROMBIN (Factor II) GENE MUTATION

- **“Prothrombin G20210A”**
- **Single nucleotide change (G20210A) in prothrombin gene - associated with hypercoagulability - leads to elevated prothrombin levels**
- **1% to 2% of population**
- **Leads to elevated levels of Prothrombin** which can be converted to **thrombin** (proteolytically cleaves fibrinogen to fibrin which then crosslinks to form fibrin clot)
- **3x increase in Relative Risk** of venous thrombosis

ELEVATED HOMOCYSTEINE (ENDOTHELIAL DAMAGE)

- Homocysteine: amino acid metabolite, **contributes to atherosclerosis as well as arterial and venous thrombosis**
- Used as biomarker to predict cardiovascular events
- Prothrombotic effects possibly due to thioester linkages formed between metabolites of homocysteine and proteins like fibrinogen
- Inherited or acquired:
 - Marked elevations of homocysteine may be caused by **inherited deficiency of cystathione β -synthetase**
 - Acquired causes include **deficiency of vitamin B 6, B12, and folic acid**

Metabolism of homocysteine consists of Vitamin B6 dependent transsulfuration pathway involving cystathionine beta-synthetase (CBS), and a Folate and Vitamin B12 dependent remethylation pathway with methylenetetrahydrofolatereductase (MTHFR) and methioninesynthase (MS)

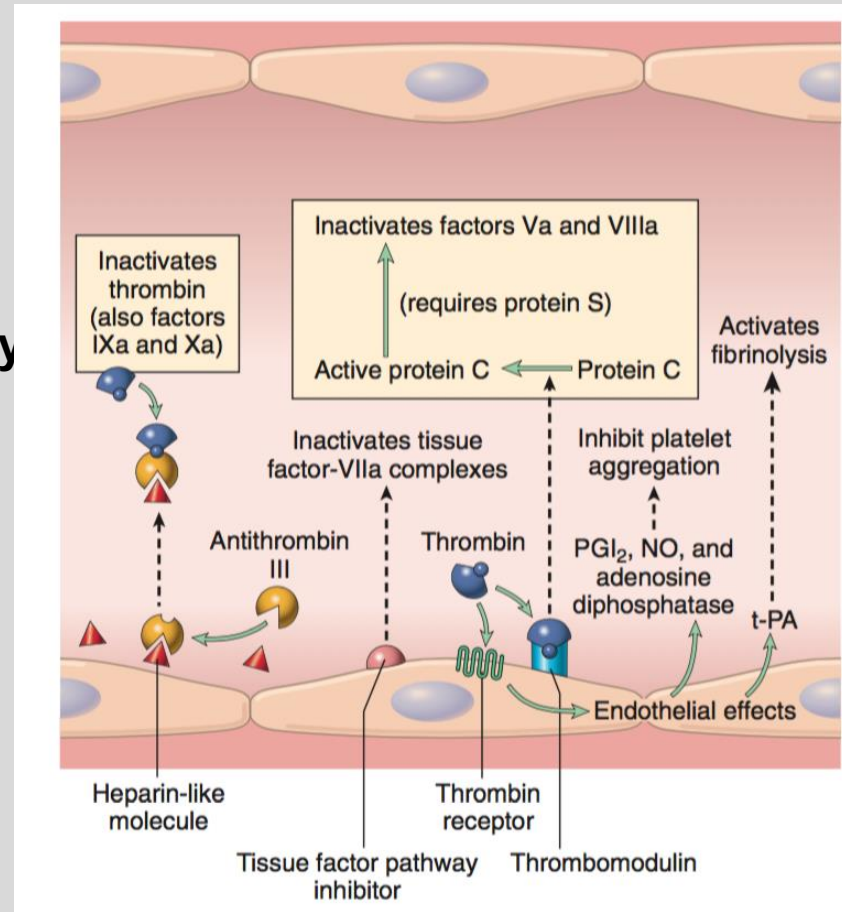


Homocysteinemia

- Lack of B12 or folate leads to decreased conversion of homocysteine to methionine; resulting in build-up of homocysteine
- Other causes: renal failure, DM, hypothyroidism, CA, Pernicious anemia, Inflammatory bowel disease or therapy with MTX, theophylline, or phenytoin

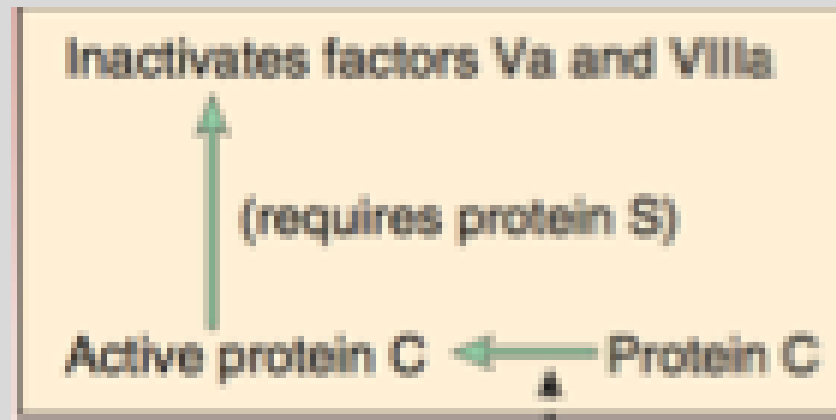
Antithrombin III Deficiency

- **Function:** Inhibits thrombin and factor X
- **Helps regulate clot formation and prevents emboli**
- **Also enhanced by HEPARIN**
- **Deficiency leads to Increased Thrombin**
- **Individuals with antithrombin III deficiency have insensitivity to heparin**
- **Rare; can be genetic (AD) > (AR);**
- **Type I: not enough made by liver; Type II: defective protein**
- **Acquired (liver disease or **nephrotic syndrome**)**
- **Mostly presents as venous thrombosis**
- **Therapeutics: direct thrombin or Xa inhibitors, warfarin, supplemental antithrombin III**



Protein C & S Deficiency

- Natural anticoagulants; made by liver, need to be activated
 - Vitamin K-dependent proteins that serves as a cofactor for activated protein C to inactivate the procoagulant factors Va and VIIIa, thereby reducing thrombin generation
 - **Mutation results in inability to inactivate factors V & VIII; results in increased thrombin formation**
 - Individuals have increased risk of thrombotic skin necrosis with warfarin
- CLINICAL CORRELATE: if patient is placed on warfarin therapy, the liver will not produce factors 2,7,9,10, protein C and S. All pre-existing protein C and S will degrade first leaving pre-existing factors 2,7,9,10 to be relatively high
- Need Heparin bridge
 - Presents with venous thrombosis and recurrent thromboembolism beginning in adolescence/early adulthood



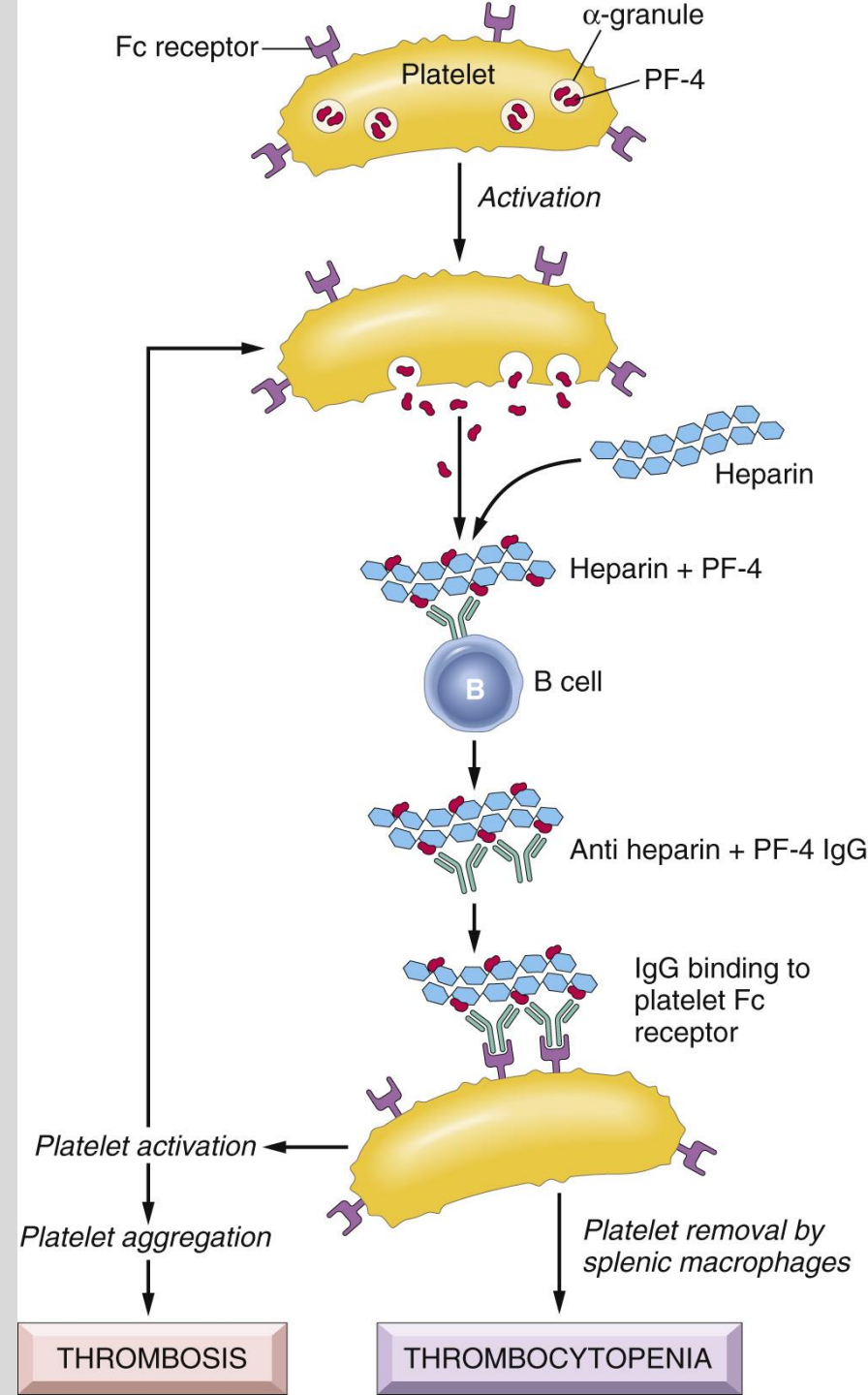
Secondary (Acquired) Hypercoagulable (Clotting) Disorders

- Multifactorial
- Prolonged bed rest
- MI
- Atrial fibrillation
- DIC
- Post-op
- Prosthetic cardiac valves
- Malignancy (disseminated) – release of prothrombotic mediators
- Smoking or obesity
- Oral Contraceptives / Pregnancy – Estrogen increases coagulation factors and decreases ATIII
- Advanced Age – Decreased endothelial PGI₂ production
- Hyperviscosity – Polycythemia / Waldenstroms macroglobulinemia
- **Heparin Induced Thrombocytopenia (HIT)**
- **Anti-Phospholipid Antibody Syndrome (APLS)**

Heparin-Induced Thrombocytopenia (HIT) Syndrome

- **Thrombocytopenia** most common manifestation (about 5% of persons receiving unfractionated heparin (see Path of Hemorrhage lecture); **thrombosis** most serious complication occurring in ~50% of cases; affects both veins and arteries
- Diagnosis: demonstration of anti-PF4-heparin antibodies
- Low-molecular-weight heparin preparations induce HIT less frequently
- May present with sudden onset of pain, redness, and swelling of an arm or leg; ecchymosis, necrosis of skin, gangrene, stroke or MI may develop
- 2 Types:
- HIT Type I: (non-immune) asymptomatic transient drop of platelets, returns to normal
- HIT Type II: (immune mediated) Clinically serious condition
- Thrombosis may occur in either veins or arteries

- Formation of antibodies that recognize complexes of heparin and PF4 on surface of platelets
- Released PF4 (+) binds to heparin (-), undergoes conformational change - **results in formation of neoantigen against which IgG antibodies are formed; IgG produced against heparin-PF4 complex**
- PF4-IgG immune complex, attaches to and cross-links Fc receptors on platelet surface - **leads to platelet activation and aggregation**
- Prothrombotic state may also be increased by activation of endothelium by binding of HIT antibodies to PF4-like proteins on their surface
- **Binding of HIT antibodies to platelets results in removal by macrophages (thrombocytopenia)**



Antiphospholipid Antibody Syndrome (APS)

- Autoimmune disorder characterized by:
 - Presence of one or more antiphospholipid (aPL) autoantibodies
 - Venous or arterial thromboses, or pregnancy complications (e.g., recurrent miscarriages, unexplained fetal death, and premature birth.)
- Primary or secondary (e.g., **lupus**)
- Primary APS: patients only have hypercoagulable state and no evidence of autoimmune diseases
- Presence of Antiphospholipid autoantibodies
 - Anticardiolipin antibodies (IgG or IgM)
 - Anti-beta-2-glycoprotein-I antibodies (IgG or IgM)
 - Lupus anticoagulants
- Various clinical manifestations: **recurrent thromboses, repeated miscarriages, cardiac valve vegetations, and thrombocytopenia**
- Clinical presentations can include **pulmonary embolism (following lower extremity venous thrombosis), pulmonary hypertension (from recurrent subclinical pulmonary emboli), valvular heart disease, stroke, bowel infarction, or renovascular hypertension**

Robbins and Cotran, Pathologic Basis of Disease, 10th ed., 2020, Ch.4

Antiphospholipid Syndrome - Pathogenesis

- Not fully understood: **APL antibodies directed against anionic membrane phospholipids or proteins associated with phospholipids** (include cardiolipin and β 2 -glycoprotein I)
- **Anti- β 2 -glycoprotein antibodies suspected to have major role in APS by activating endothelial cells, monocytes, and platelets**
- Patients with APS also show evidence of complement activation and inhibition of fibrinolytic processes: both favor **prothrombotic state**
- Pregnancy morbidity: **defining manifestation of APS; fetal loss not thought to be caused by thrombosis, but from antibody-mediated interference with growth and differentiation of trophoblasts: leads to failure of placentation**

APL Antibodies

- Antiphospholipid antibodies also identified in 5% to 15% of normal individuals, implying Abs not sufficient to cause full-blown syndrome
- Antibodies induce hypercoagulable state in vivo- but interfere with phospholipids and inhibit coagulation in vitro, **prolonging the PTT**
- Mixing study: patient plasma mixed with control plasma does not correct aPTT
- Often **false-positive serologic test for syphilis** because the antigen in standard assay is embedded in cardiolipin, which cross reacts with phospholipids of *Treponema pallidum*
- Diagnosis based on clinical features and demonstration of APL antibodies in serum
- Therapy - various forms of anticoagulation

Robbins and Cotran Pathologic Basis of Disease, 10th ed., 2021

Antiphospholipid Syndrome (APS)

**Underlying Lupus
("Secondary APS")**

**Infection, medication, malignancy,
other chronic disease**

Idiopathic

Development of certain anti-phospholipid antibodies (aPL): anti-cardiolipin antibodies, anti- β 2-glycoprotein-I antibodies, and/or lupus-type inhibitors (aka "lupus anticoagulants")

Note: Lupus-type
Inhibitors interact
with a phospholipid
used in lab testing

**↑ PTT *in vitro*,
normal PTT *in vivo***

Author:
Adam Burgess
Reviewers:
Haotian Wang
Alexander Arnold
Lynn Savoie*
* MD at time of publication

Inhibition of proteins
C & S, antithrombin,
and tPA, and ↑
platelet activation
and endothelial
expression of tissue
factor, all via
antibody binding

**Hypercoagulable state
(↑ risk of thrombosis)**

and

Complement
activation and cytokine
dysregulation (e.g.
↑ TNF- α)

Proinflammatory state

and

Direct interactions of
antibodies with tissues

Pathogenesis of complications is likely a mix of these factors

Fetal Loss

Placental thrombosis, complement activation,
direct inhibition of the trophoblast

Fetal loss

Thromboses

Venous thrombosis

DVT, PE

Arterial thrombosis

Stroke, TIA

Arterial or venous or micro- thromboses

Nephropathy

Micro-thromboses

Skin ulcerations

Livedo reticularis

Other thromboses (ocular, coronary, adrenal, Budd-Chiari, osteonecrosis)

Other findings and complications

Micro-thromboses, direct antibody
interactions in the brain

**Cognitive changes,
epilepsy, migraine**

Direct antibody interactions on heart valves

Heart valve thickening

Development of anti-prothrombin antibodies

Bleeding (uncommon)

↑ platelet destruction by RE system

Thrombocytopenia

AIHA and/or MAHA

Hemolytic Anemia

Positive antibody test

+

**History of significant
thrombosis or fetal loss**

Diagnosis of APS

Abbreviations:

tPA – Tissue Plasminogen
Activator
AIHA – Autoimmune
Hemolytic Anemia
MAHA – Microangiopathic
Hemolytic Anemia
RE – Reticuloendothelial
DVT – Deep Vein
Thrombosis
PE – Pulmonary Embolism
TIA – Transient Ischemic
Attack

Other Secondary Causes

- **Prolonged Bed Rest:** reduced venous return from the legs leads to stasis
- **Oral Contraceptives** or Hyperestrogenic state from **pregnancy** increase hepatic synthesis of coagulation factors and reduce anticoagulant synthesis
- **Myocardial Infarction:** area of infarct has poor contracting heart muscle leading to stasis
- **Atrial Fibrillation:** stasis of blood in the left atrial appendage
- **Malignancy:** tumors release procoagulants into the blood stream
- **Hyperviscosity** (such as polycythemia vera): high concentrations of prothrombotic microparticles, stasis, and activation of platelets, leukocytes, and endothelial cells; stasis
- **Old age:** reduced endothelial PGI 2 production
- **Post-Op:** vascular damage and prolonged bed rest
- **Prosthetic Cardiac Valves**

Adapted Osmosis/USMLE example question

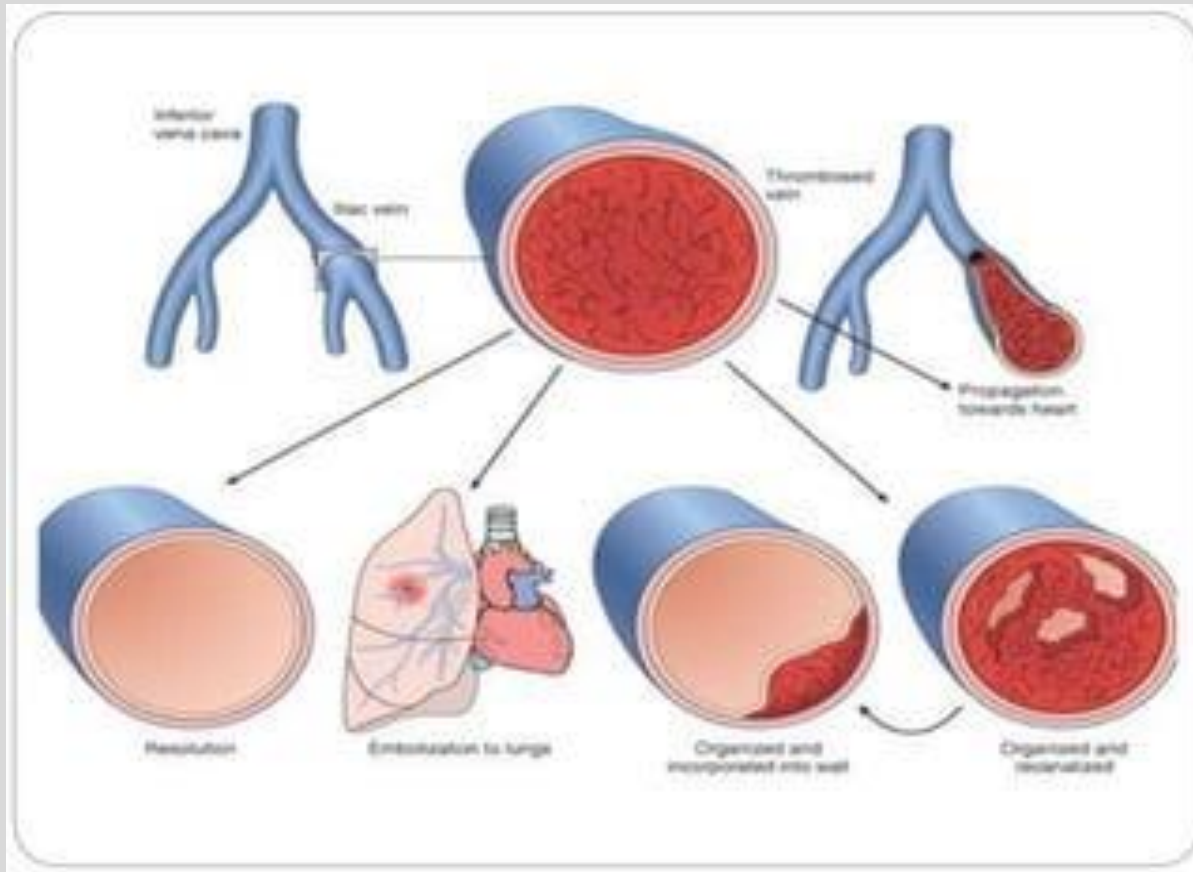
- A 25-year-old Caucasian woman presents to the ED due to difficulty breathing. Two hours ago, she suddenly felt chest pain accompanied by dyspnea. The patient denies any recent trauma, and she does not smoke or use illicit drugs. She is on no meds including contraception pills. She is afebrile. On physical examination, the patient appears distressed. Heart sounds are normal and lungs are clear to auscultation. Labs show an elevated d-dimer. PT and PTT are 12 (normal) and 10 seconds (decreased), respectively. Purified protein C is added to the patient's plasma and causes no change in the lab values. Which of the following is the most likely diagnosis?
- **A. Protein S deficiency**
- **B. Prothrombin gene mutation**
- **C. Antiphospholipid syndrome**
- **D. Antithrombin deficiency**
- **E. Factor V Leiden deficiency**

INCORRECT ANSWERS EXPLANATIONS

- A. Protein S deficiency: Although protein S deficiency is difficult to distinguish from factor V Leiden deficiency, the former is far rarer and therefore less likely.
- B. Prothrombin gene mutation: Prothrombin, also known as factor II, is the precursor of thrombin, the end product of the coagulation cascade. Thrombin's main role is to proteolytically cleave fibrinogen to fibrin, which in turn crosslinks to form a fibrin clot. It also interacts with other hemostatic components. The G20210A point mutation in the prothrombin gene results in a gain-of-function of prothrombin that leads to hypercoagulability. However, the failure to change PTT after adding purified protein C favors another diagnosis.
- C. Antiphospholipid syndrome: Autoimmune condition characterized by venous and/or arterial thrombosis due to the presence of antiphospholipid antibodies (lupus anticoagulant, anticardiolipin, anti- β 2 glycoprotein I antibodies). It is most commonly associated with systemic lupus erythematosus (SLE). Patients with APS often show prolonged PTT. In contrast, this patient shows no features of SLE and has reduced PTT.
- D. Antithrombin deficiency: Antithrombin (AT, previously called AT III) deficiency may be acquired or inherited in an autosomal dominant fashion with variable penetration. Antithrombin is a natural anticoagulant that inhibits thrombin (factor IIa), factor Xa, and other factors in the coagulation cascade. A variety of mutations have been recognized. AT has no direct effect on the PT, PTT, or thrombin time. Instead, it diminishes the increase in PTT following heparin administration. This phenomenon exists since heparin, an indirect inhibitor of thrombin and factor Xa, requires functional AT to be effective.

FATE OF THROMBUS

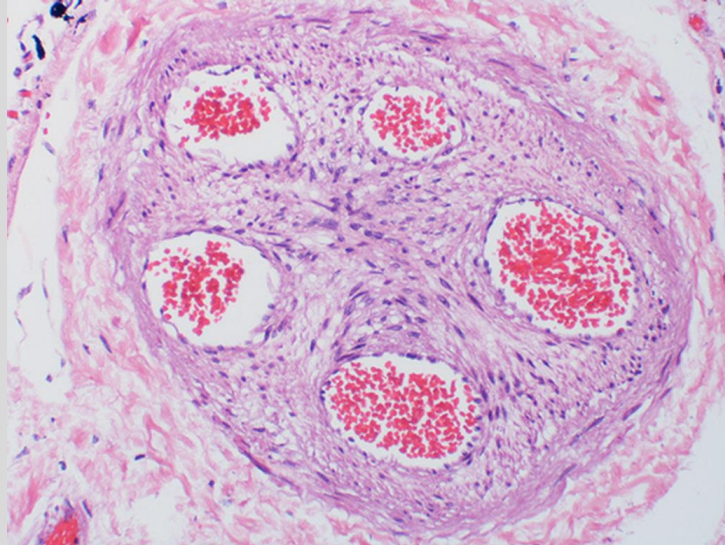
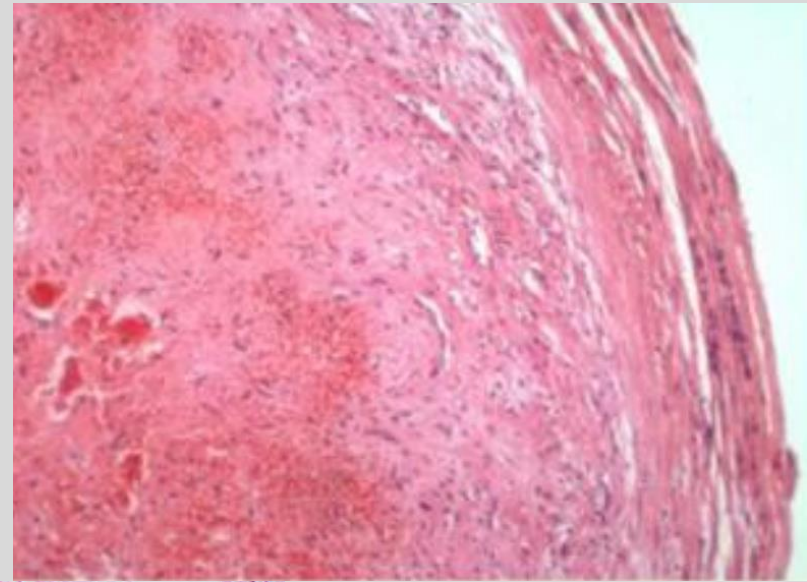
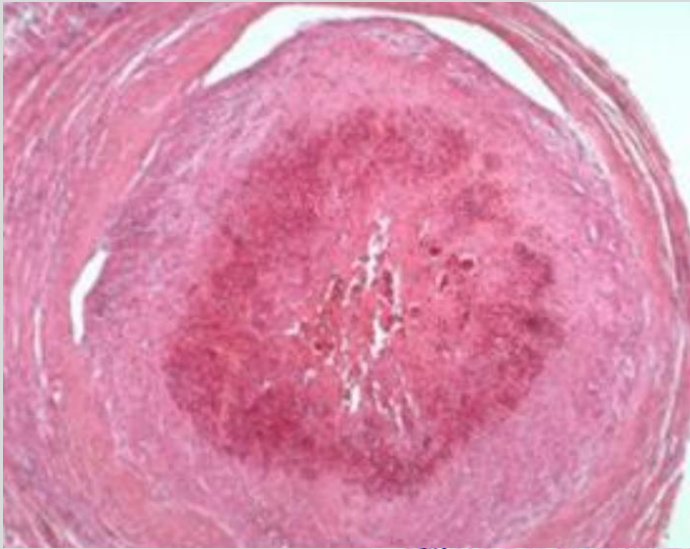
- **Propagation:** additional platelets/fibrin attach and increase the size of thrombus
- **Embolization:** thrombi dislodge, travel to other sites in vasculature
- **Dissolution/Resolution:** result of fibrinolysis, which can lead to rapid shrinkage and total disappearance of recent thrombi
 - extensive fibrin deposition and cross-linking in **older thrombi: more resistant to lysis** (therapeutic administration of fibrinolytic agents such as t-PA (e.g., in acute coronary thrombosis) usually effective only when given during first few hours of thrombotic event
 - **Organization and recanalization:** older thrombi become organized by ingrowth of endothelial cells, smooth muscle cells, and fibroblasts; capillary channels eventually form within thrombus which allows blood flow



RESOLUTION EMBOLISM ORGANIZATION/
INCORPORATION
INTO VESSEL WALL RECANALIZATION

<https://www.slideshare.net/VarugheseGeorge/thrombosis-embolism-and-infarction>

Recanalization



Embolism

Embolus - Detached intravascular **solid, liquid, or gaseous mass** carried by blood **from point of origin to distant site**, often causing tissue dysfunction or infarction

- Vast majority are dislodged thrombi (**thromboembolism**)
- Other rare emboli - fat droplets, nitrogen bubbles, atherosclerotic debris (cholesterol emboli), tumor fragments, bone marrow, or foreign bodies

Emboli travel through blood, reach vessels too small to pass through, cause partial or complete vascular occlusion

Forms of Emboli

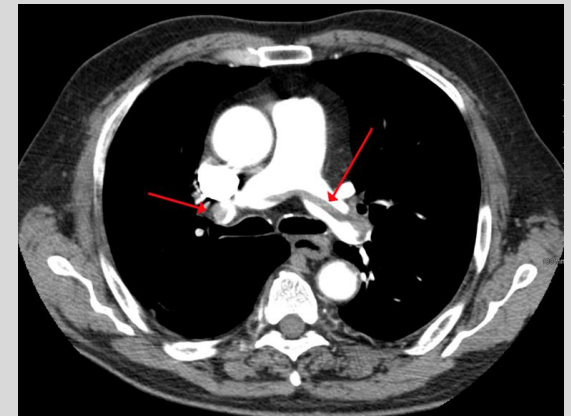
- Thrombotic (***Lines of Zahn***) - *thromboembolism*
- Atherosclerotic (***Cholesterol Clefts***)
- Fat and Bone Marrow (***Fat Droplets & Marrow***)
- Air or Gas
- Amniotic Fluid (***Squamous Cells Swirls and Keratin Debris***)

Pulmonary Thromboembolism

- Disruption of blood flow to pulmonary artery or its branches usually originating from a dislodge DVT of the lower extremities most often above the popliteal fossa (95%)
- Symptoms include dyspnea, pleuritic chest pain, cough, hemoptysis, presyncope, or syncope

Pulmonary Embolism

- **Originate from DVT (95%);** most common form of thromboembolic disease
- Fragmented thrombi travel through progressively larger veins to right heart and “slam” into pulmonary vasculature; depending on embolus size **can:**
 - **occlude main pulmonary artery**
 - **straddle pulmonary artery bifurcation (saddle embolus), or**
 - **pass into smaller, branching arteries**
- Often multiple emboli, occur sequentially or simultaneously, as shower of smaller emboli from single large mass
- Most pulmonary emboli (60% to 80%) small and clinically silent



https://twitter.com/hp_ems/status/479944912675143680

Robbins & Cotran Pathologic Basis of Disease, 10th ed., 2020 – Chapter 14

Major Consequences of PE

- **When emboli obstruct 60% or more of pulmonary circulation: sudden death, acute right heart failure (cor pulmonale), or cardiovascular collapse**
- **Embolic obstruction of medium-sized arteries with subsequent vascular rupture: can result in pulmonary hemorrhage – usually NOT pulmonary infarction** (note: lung supplied by both pulmonary arteries and bronchial arteries; intact bronchial circulation can usually sufficiently perfuse affected area)
- Embolic obstruction of **small end-arteriolar** pulmonary branches: often results in hemorrhage or infarction
- **Multiple emboli** over time may cause **pulmonary hypertension** and right ventricular failure

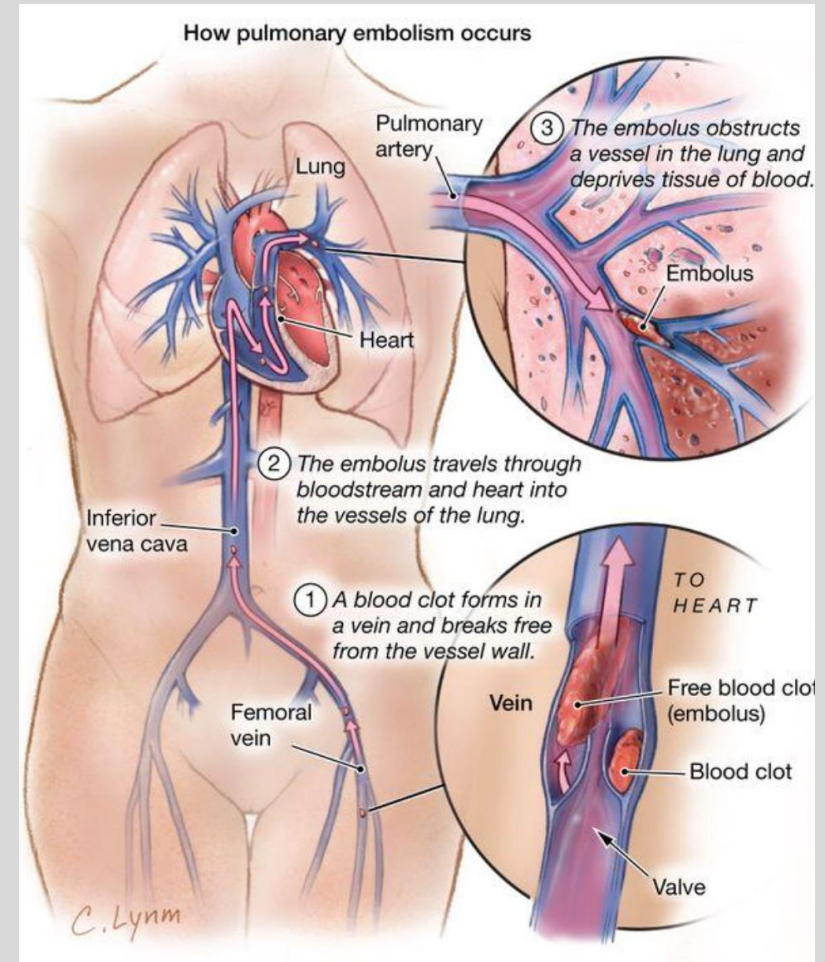
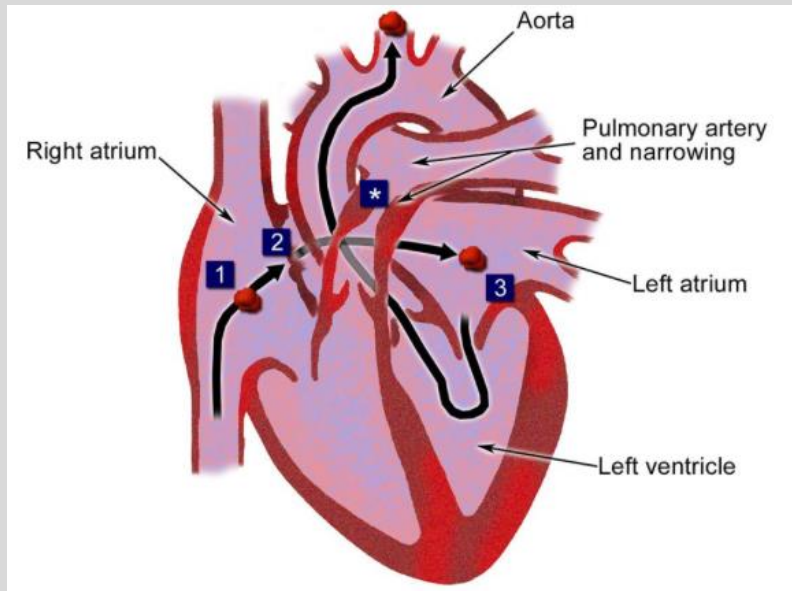
Systemic Thromboembolism

- **Arise in or affect arterial system**
- **Most systemic emboli (80%) from intracardiac mural thrombi**
 - 2/3rds associated with left ventricular wall infarcts (stasis and endothelial damage)
 - 1/4th with left atrial dilation and fibrillation (stasis)
 - remainder originate from aortic aneurysms, atherosclerotic plaques, valvular vegetations, or venous thrombi (paradoxical emboli)
 - 10% to 15% unknown origin
- Arterial emboli can travel to many sites (**v. venous thrombi, majority travels to lung**); most go to lower extremities (75%) or brain (10%); other tissues (e.g., intestines, kidneys, spleen, and upper extremities) may be involved
- Usually result in tissue infarction

Embolism

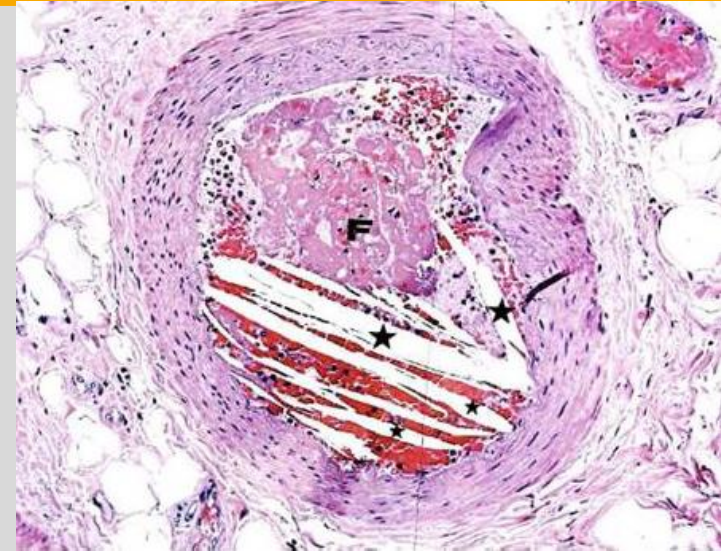
Paradoxical Emboli

- Rare; venous embolus passes into arterial system via an atrial or ventricular **septum defect**



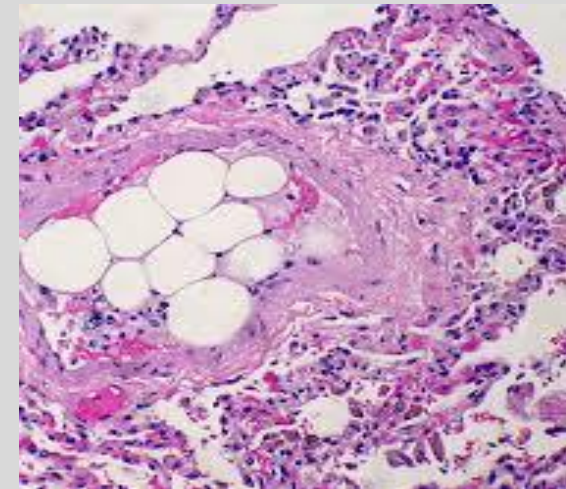
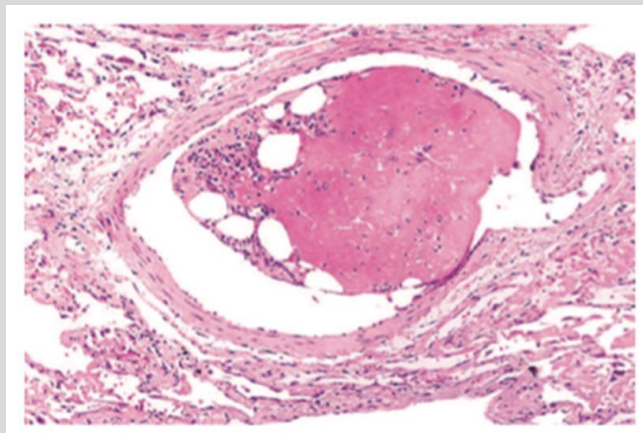
Atherosclerotic Emboli (atheroemboli)

- Occurs when atheromatous plaques in large arteries release cholesterol emboli and atheromatous debris into circulation
- Often occludes small caliber vessels
- Frequently occurs as a result of intraarterial procedures
- Histology shows CHOLESTEROL CLEFTS (crystals) surrounded by hyperplastic intimal tissue and giant cells
- Most commonly affects kidney, skin, and GI tract



FAT/BONE MARROW EMBOLISM

- Presence of microscopic fat globules (sometimes with associated hematopoietic bone marrow) in vasculature after **fractures of long bones** - fat in bone marrow released into vascular sinusoids
- Fat microemboli and red cell and platelet aggregates **occlude pulmonary and cerebral microvasculature**
- Pathogenesis
 - **Mechanical obstruction of microvasculature**; RBCs and platelets aggregated by fat may obstruct pulmonary and cerebral microvasculature
 - **Biochemical injury** due to toxicity to endothelium by free fatty acids; also activates platelets and granulocytes



FAT EMBOLISM SYNDROME

- Symptomatic patients (minority) - **pulmonary insufficiency, neurologic symptoms, anemia, thrombocytopenia**
- **Diffuse petechial rash; fatal in 5% to 15% of cases**
- **Thrombocytopenia** because of platelet adhesion to fat globules and subsequent aggregation or splenic sequestration
- **Anemia** can result from red cell aggregation and/or hemolysis
- Usually, 1 to 3 days after injury - sudden onset of tachypnea, dyspnea, and tachycardia; irritability; restlessness, can progress to delirium or coma

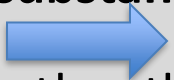
Air Embolism

- Occurs when communication between vasculature and outside air exists; negative pressure gradient “sucks” in air
- **Air may be introduced during endovascular and invasive procedures, mechanical ventilation;** large volume of air (**more than 100 mL**) needed to cause effect in pulmonary circulation
- Introduction of 300 to 500 mL of air at 100 mL/sec may be fatal
- Coronary Bypass Surgery, Laparoscopic Abdominal or Obstetric Surgery, Central Line Placement, Chest wall injuries, Scuba & Deep sea divers, Underwater Construction
- **Air in pulmonary vasculature blocks perfusion of downstream region; and microemboli of air trapped in pulmonary capillaries cause intense inflammatory response with release of cytokines that can injure alveoli**
- Bubbles in central nervous system can cause mental impairment and sudden onset of coma
- **Classic presentation: sudden collapse, neurologic symptoms that mimic stroke, apnea, and even cardiovascular collapse**

Decompression sickness

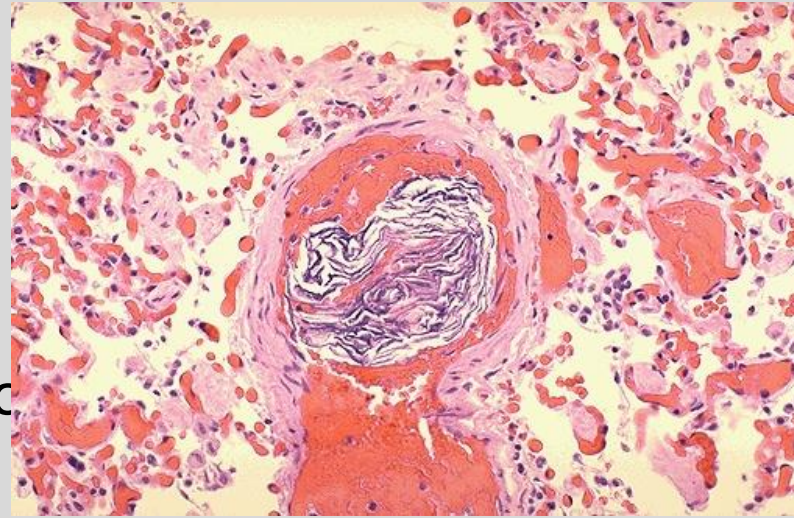
- Form of gas embolism - patients experience sudden decreases in atmospheric pressure
- Scuba, deep sea divers, and underwater construction workers
- When air is breathed at high pressure (e.g., deep sea dive), increased amounts of gas (especially nitrogen) dissolve in blood and tissues; if diver **ascends (depressurizes)** too rapidly, **nitrogen comes out of solution in tissues and blood**
- **Rapid formation of gas bubbles within skeletal muscles and supporting tissues in/ around joints: causes pain (the bends)**
- **In lungs, gas bubbles in vasculature cause edema, hemorrhage, and focal atelectasis or emphysema; respiratory distress (the chokes)**
- Chronic form of decompression sickness: **Caisson disease: persistence of gas emboli in skeletal system leading to many foci of ischemic necrosis**; more common sites: femoral heads, tibia, and humeri
- Acute decompression sickness: treated by putting patient in a chamber under enough high pressure to force gas bubbles back into solution; subsequent slow decompression allows gradual resorption and exhalation of gases, preventing obstructive bubbles from reforming

Amniotic Fluid Embolism

- Fifth most common cause of maternal mortality worldwide; about 10% of maternal deaths in US; results in permanent neurologic deficit in up to 85% of survivors
 - Incidence approximately 2 to 6 in 100,000 deliveries; mortality rate up to 80%
 - **Onset: sudden severe dyspnea, cyanosis, and shock, followed by neurologic impairment (ranging from headache to seizures and coma), and DIC**
 - Morbidity/mortality mostly from:
 - **biochemical activation of coagulation factors**
 - **components of the innate immune system, and**
 - **release of vasoactive substances** that cause INTENSE VASOCONSTRICTION in pulmonary circulation  pulmonary hypertension and right heart failure as well as DIC (rather than obstruction)
-  leads to **hypoxia, left heart failure, pulmonary edema, and diffuse alveolar damage**

Amniotic Fluid Embolism

- Underlying cause: infusion of amniotic fluid or fetal tissue into maternal circulation via tear in placental membranes or rupture of uterine veins
- Caused from disruption of placenta-amniotic interface and entry of amniotic fluid and fetal elements into maternal circulation
- Autopsy findings, micro: presence of squamous cells shed from fetal skin, lanugo hair, fat from vernix caseosa, and mucin from fetal respiratory or GI tract in the maternal pulmonary microvasculature



Epithelial squames in a peripheral pulmonary artery

Summary Slide

- Morphology of thrombi – lines of Zahn v. post-mortem clots, Virchow's triad (endothelial injury, abnormal blood flow, hypercoagulable state)
- Primary thrombotic (clotting) disorders
 - Factor V Leiden – AD, most common, Factor V resistant to cleavage by protein C
 - Prothrombin gene mutation – AD, increase prothrombin, increased thrombin
 - Homocysteinemia – endothelial damage, prothrombotic effects
 - Anti-thrombin III deficiency – def. leads to increased thrombin
 - Protein C and S deficiencies – inability to inactivate factors V and VIII: results in increased thrombin
- Secondary thrombotic (clotting) disorders
 - Malignancies, OCP, pregnancy, age, hyperviscosity
 - Heparin induced thrombocytopenia – induced IgG autoAbs to heparin complexes and platelet factor 4
 - Anti-Phospholipid Ab syndrome – recurrent thrombosis, repeated miscarriages, decreased platelets
- Fate of thrombi – propagation, recanalization, embolization, dissolution
- Venous v. arterial and cardiac thrombi
- Emboli – pulmonary emboli, paradoxical emboli, systemic emboli, fat emboli, air emboli, and amniotic fluid emboli

Special thanks to Academic Scholar Alex Toufexis 2023

LECTURE FEEDBACK FORM

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

THANK YOU!